

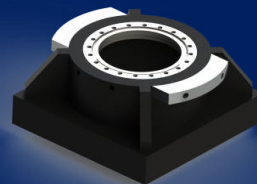
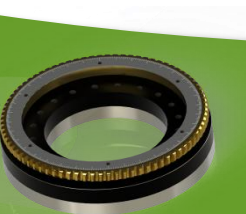
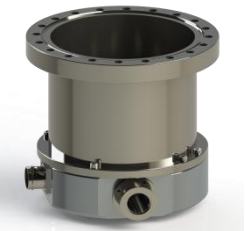


System Designs & Implementations



羅仁佑 Robert Jen-Iu Lo

- Windowless fluorescence absorption
- Photochemistry system in U9
- Absorption & Emission System In HF (03B)
- Gas Absorption System In HF (03B)
- Photodesorption, emission system
- Knudsen Cell
- Gas sample inlet
- VUV absorption system
- Microwave discharge lamp and absorption cell
- Stark effect system
- Atom gun system



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Windowless fluorescence absorption system --- temperature variation

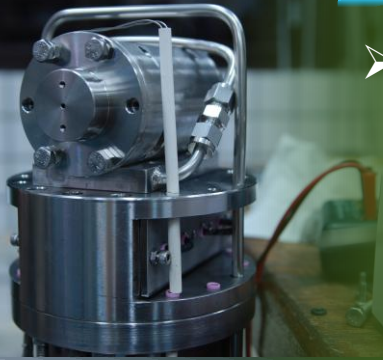


Full System Assembly



➤ System Overview

- LN cryo-head
- Spectrometer
- VUV (Differential pumping)
- VUV flux detector
- Solar blind PMT(160~320 nm)



➤ Differential pumping system

- 2 stages, 2 turbo
- Lined 3 nozzle tubes

➤ Home made spectrometer

- 5 stages MCPs
- VUV grating
- 1 turbo pump

➤ LN cryo-head

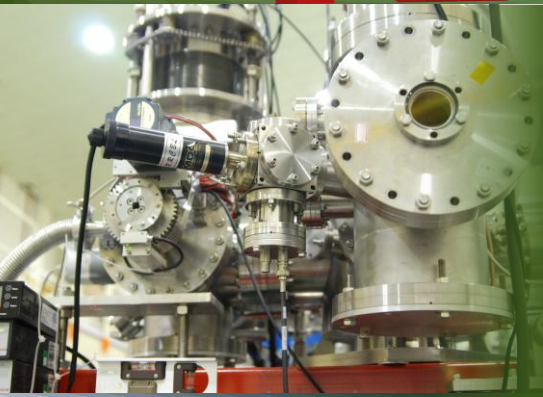
- Cell (absorption & fluorescence)
- Low temperature detector (diode 4 lines)
- heater

Windowless fluorescence absorption system --- temperature variation



Full System Assembly with main frame

- System Overview
 - LN cryo-head
 - Spectrometer
 - VUV (Differential pumping)
 - VUV flux detector
 - Solar blind PMT(160~320 nm)



➤ Differential pumping system

- 2 stages, 2 turbo
- Lined 3 nozzle tubes



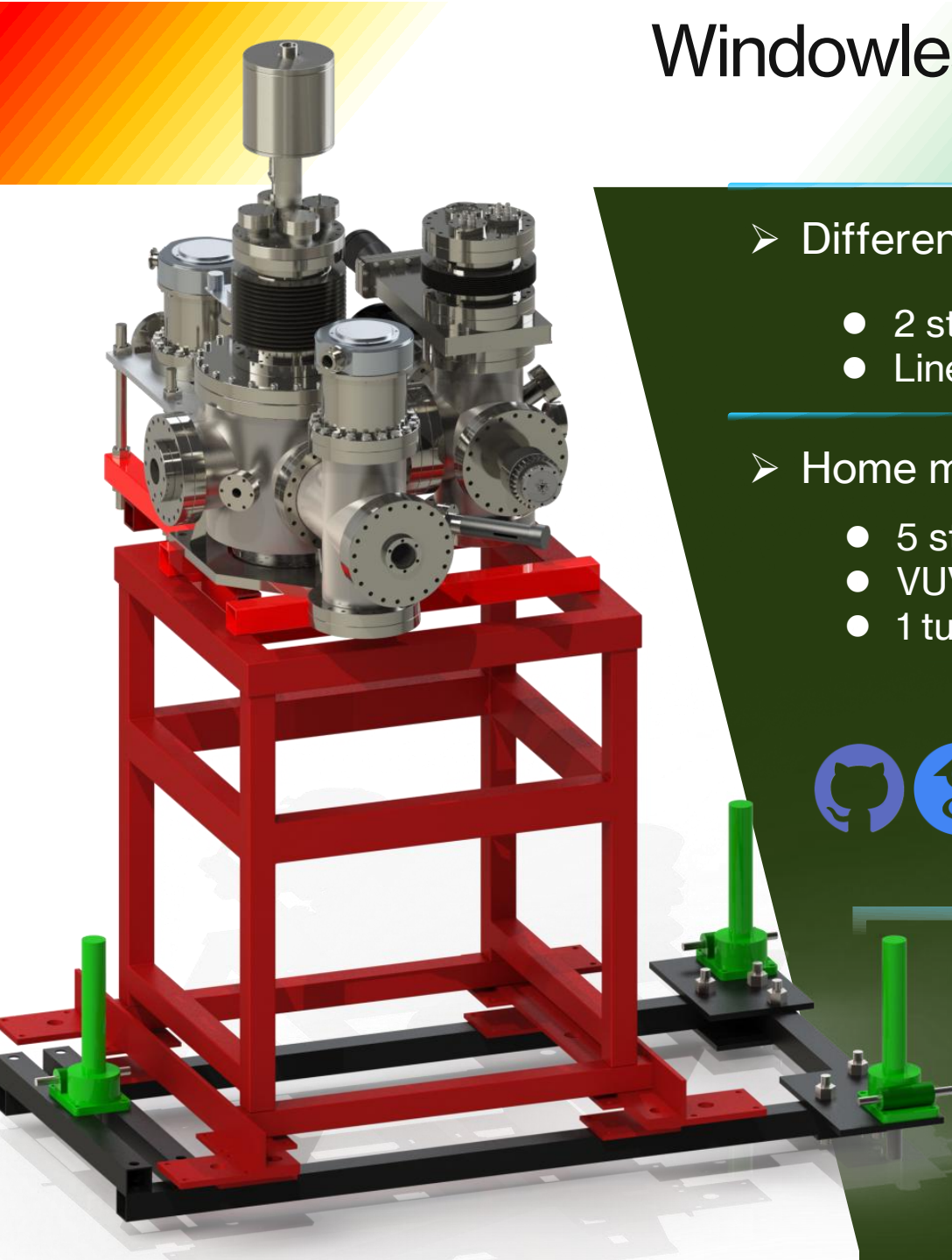
➤ Home made spectrometer

- 5 stages MCPs
- VUV grating
- 1 turbo pump

➤ LN cryo-head

- Cell (absorption & fluorescence)
- Low temperature detector (diode 4 lines)
- heater

Windowless fluorescence absorption system --- temperature variation



➤ Differential pumping system

- 2 stages, 2 turbo
- Lined 3 nozzle tubes

➤ Home made spectrometer

- 5 stages MCPs
- VUV grating
- 1 turbo pump



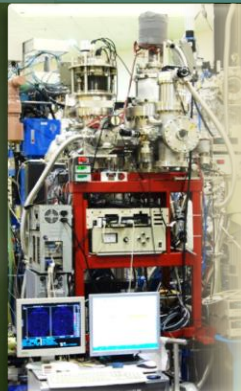
Full System Assembly extra frame for U9 (21B2)

➤ System Overview

- LN cryo-head
- Spectrometer
- VUV (Differential pumping)
- VUV flux detector
- Solar blind PMT (160~320 nm)

➤ LN cryo-head

- Cell (absorption & fluorescence)
- Low temperature detector (diode 4 lines)
- heater



Photochemistry system in U9 (21A2)



➤ Front side view

➤ System Overview

- FTIR
- iHR320
- 4K cryo-head
- Gsa inlet
- RNN vacuum rotator

➤ 4K cryo-head

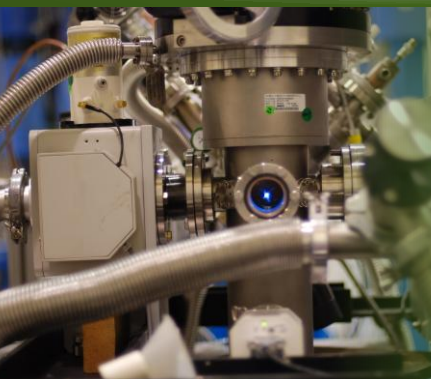
- Temp sensor (diode 4 lines)
- KBr substrate

➤ FTIR

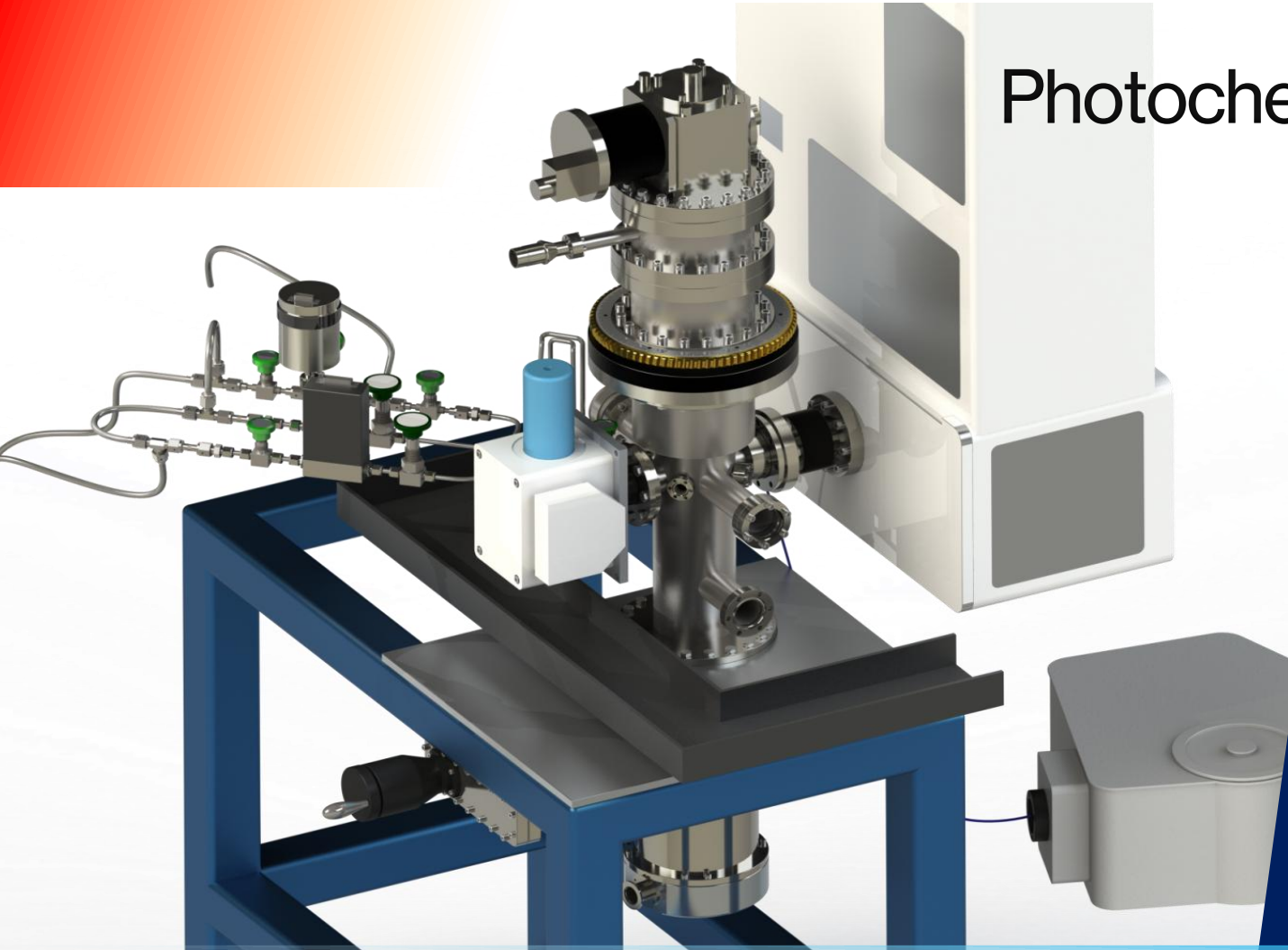
- Vacuum optical path
- IR detector

➤ iHR320

- 3 grating 600 1200 2400 l/mm
- CCD LN cooled
- Optical fiber+collimator



Photochemistry system in U9 (21A2)

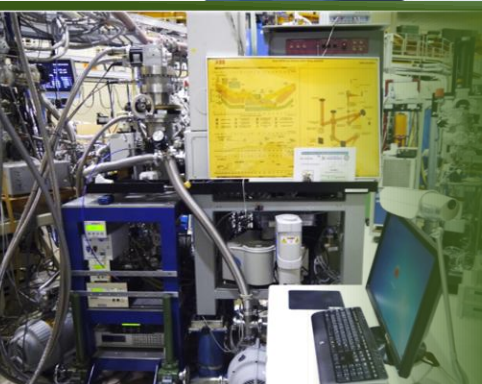


➤ End side view



➤ System Overview

- FTIR
- iHR320
- 4K cryo-head
- Gsa inlet
- RNN vacuum rotator



➤ 4K cryo-head

- Temp sensor (diode 4 lines)
- KBr substrate

➤ FTIR

- Vacuum optical path
- IR detector

➤ iHR320

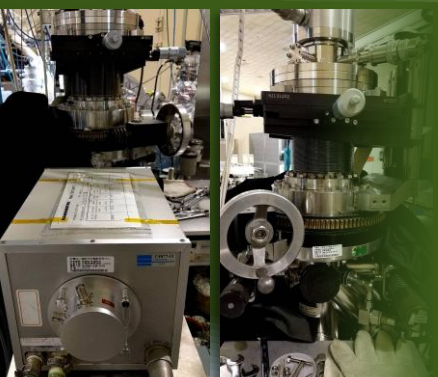
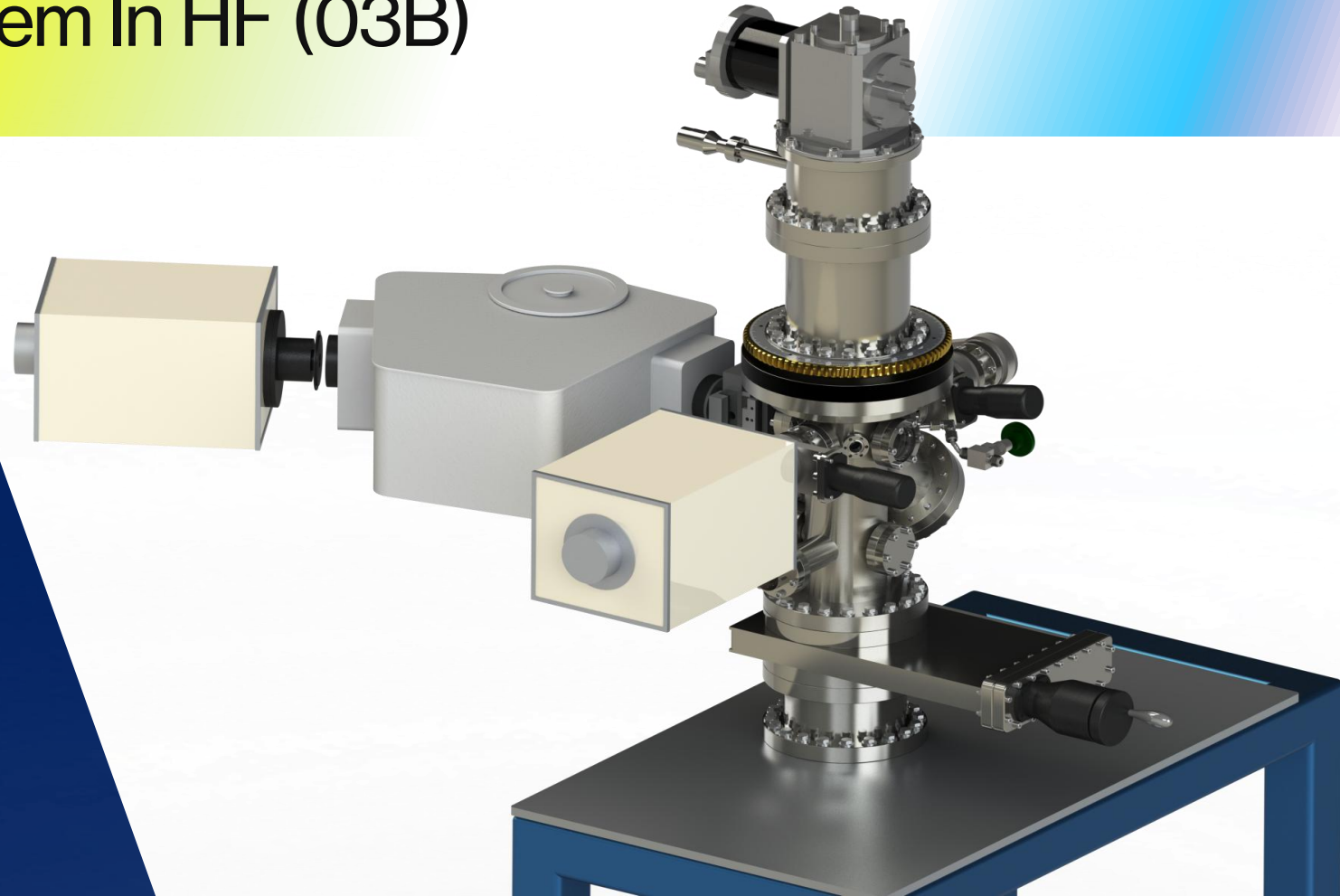
- 3 grating 600 1200 2400 l/mm
- CCD LN cooled
- Optical fiber+collimator

Absorption & Emission System In HF (03B)



➤ System Overview

- BL03 HF
- Solid state Exp.
- iHR320
- 4K cryo-head
- Gsa inlet (deposited on LiF)
- RNN vacuum rotator



➤ 4K cryo-head

- Temp sensor (diode 4 lines)
- LiF substrate

➤ NSRRC

- High Flux BL03
- Branch B

➤ iHR320

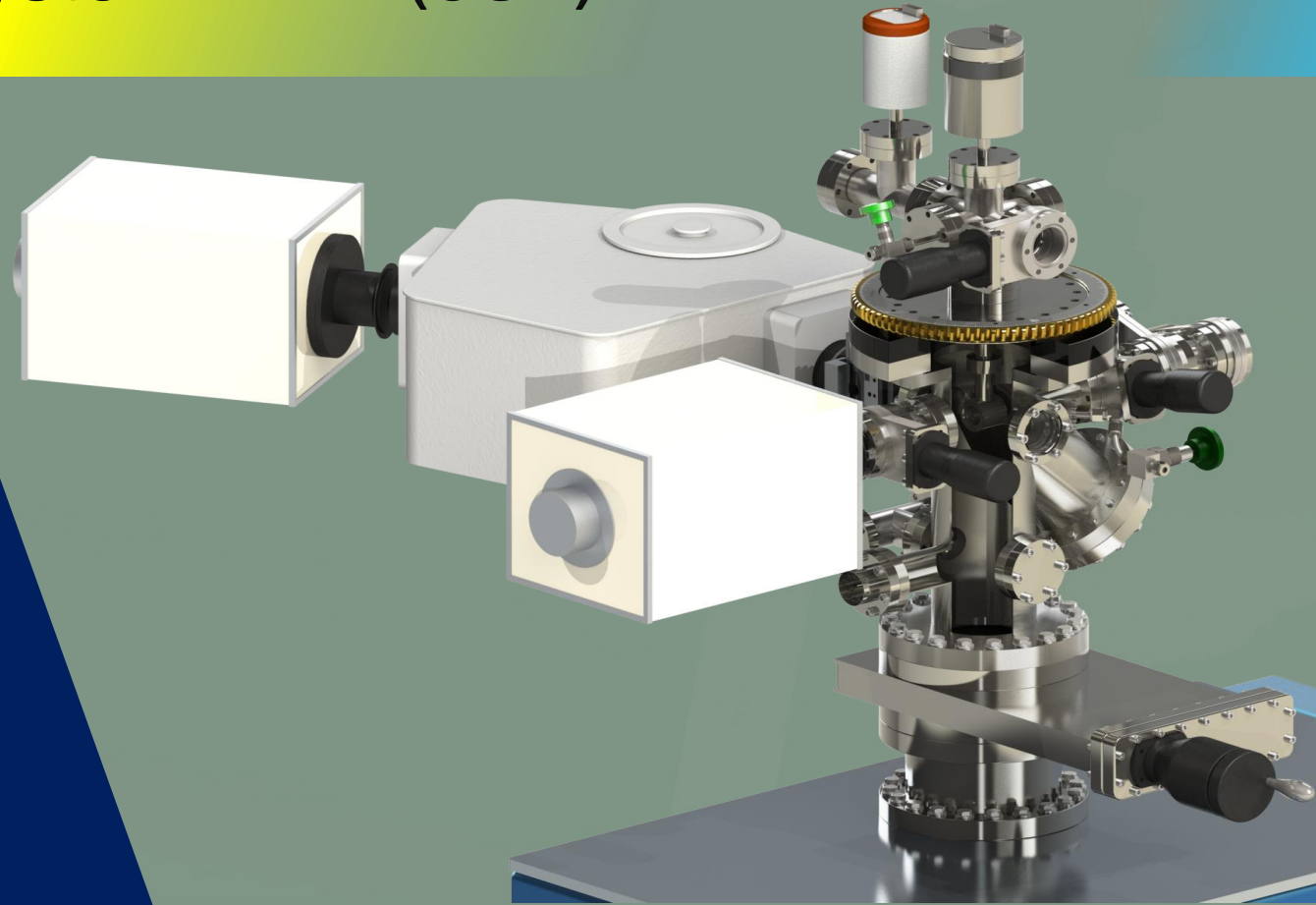
- 3 grating 600 1200 1800 l/mm
- PMT TE cooled

Gas Absorption System In HF (03B)



➤ System Overview

- BL03 HF
- Absorption cell
- iGsa inlet
- RNN vacuum rotator
- PMT TE cooling



➤ Absorption cell

- Length 10 cm
- LiF vacuum windows

➤ NSRRC

- High Flux BL03
- Branch B

➤ iHR320

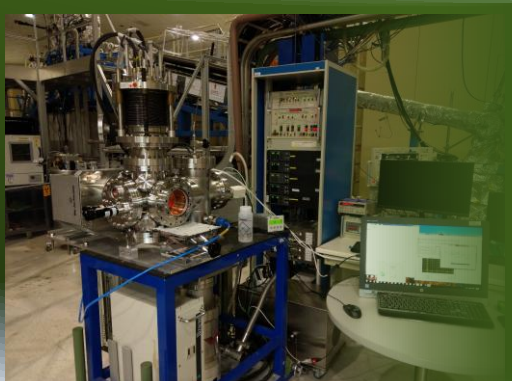
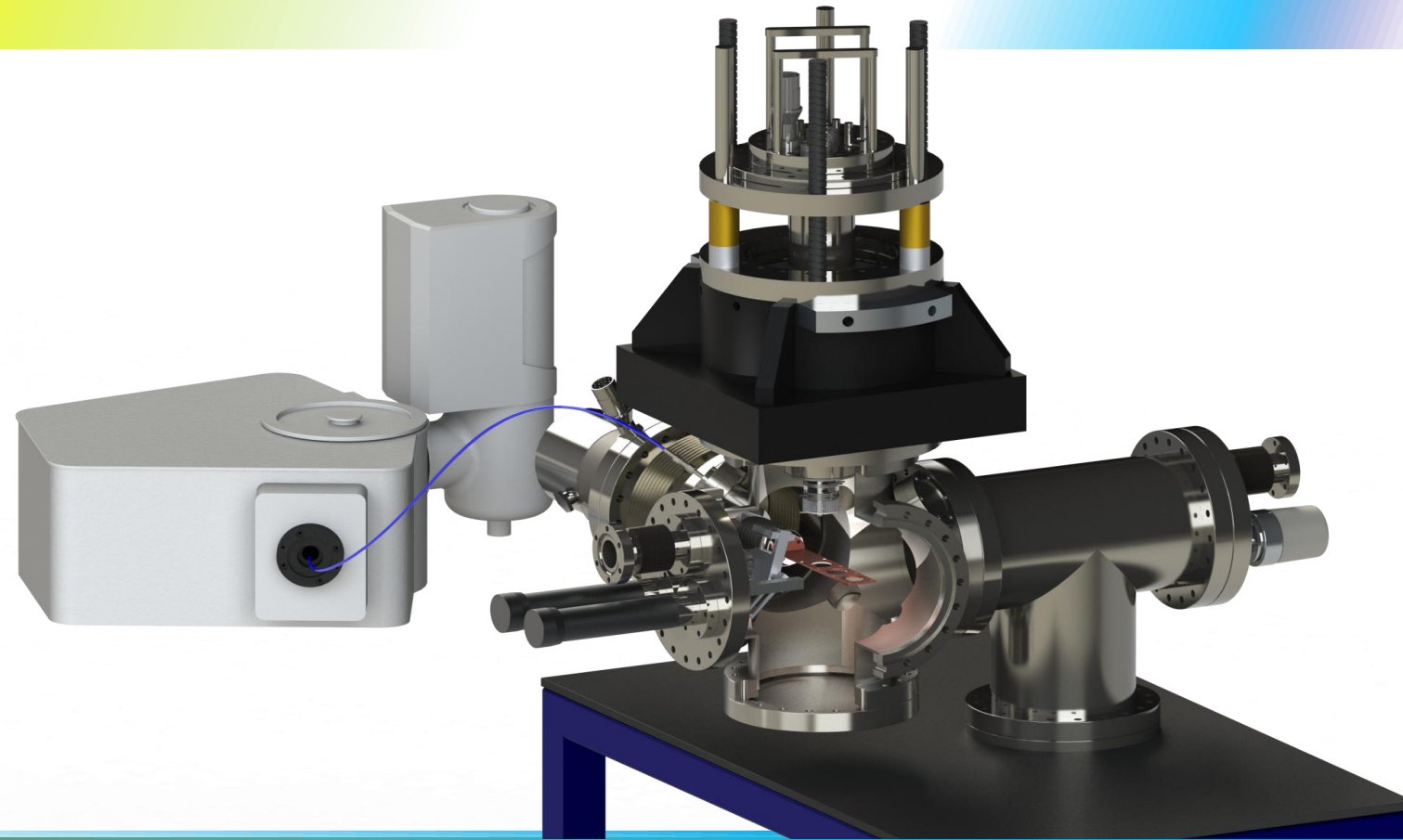
- 3 grating 600 1200 1800 l/mm
- PMT TE cooled

Photodesorption, emission system



➤ System Overview

- BL08B
- iHR320
- QMS
- vacuum translator(Mass)
- Golden mirror (reflecting VUV)



➤ Mass spectrum meter

- QMS (quadrupole)
- 200 amu
- High throughput

➤ NSRRC

- BL08
- Branch B

➤ iHR320

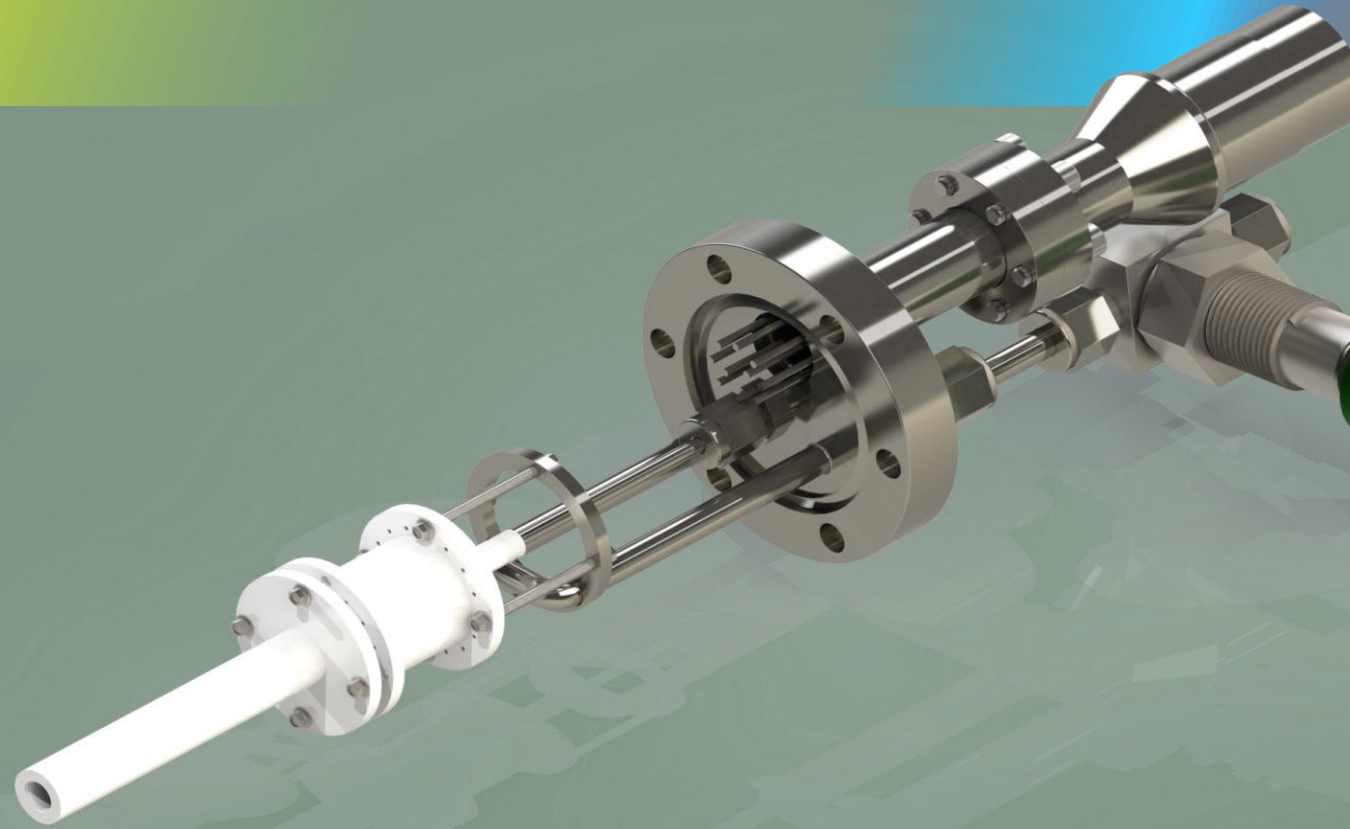
- 3 grating 600 1200 2400 l/mm
- CCD LN cooled

Knudsen Cell



➤ System Overview

- Sample oven
- Nozzle oven
- Electric feedthrough
- Buffer gas inlet
- Adjustable valve



➤ Sample oven

- 30 mm long
- Diameter 20mm

➤ Nozzle oven

- 10 cm long
- Nozzle size 4mm

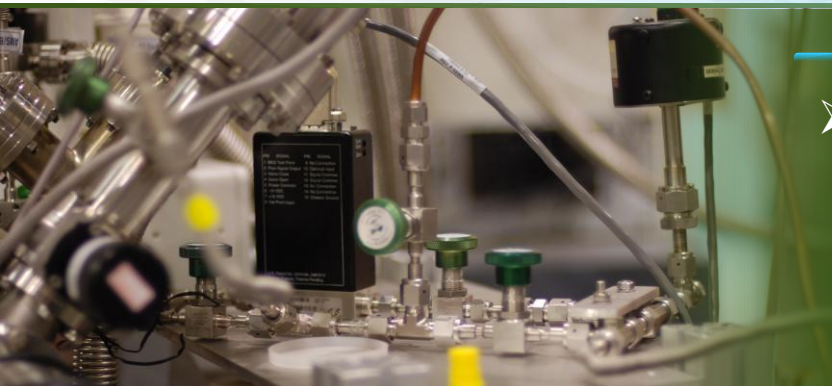
➤ Flange of holder

- 2.75 in
- 1.33 in feed through
- 12 lines feed through

Gas sample inlet system for photochemistry



- System Overview
 - 4 on-off valves
 - 2 adjustable valves
 - Flow controller
 - Baratron pressure gauge



- swagelok
 - on-off valve
 - adjustable valve

- Flow controller
 - 10 sccm
 - Hand manual

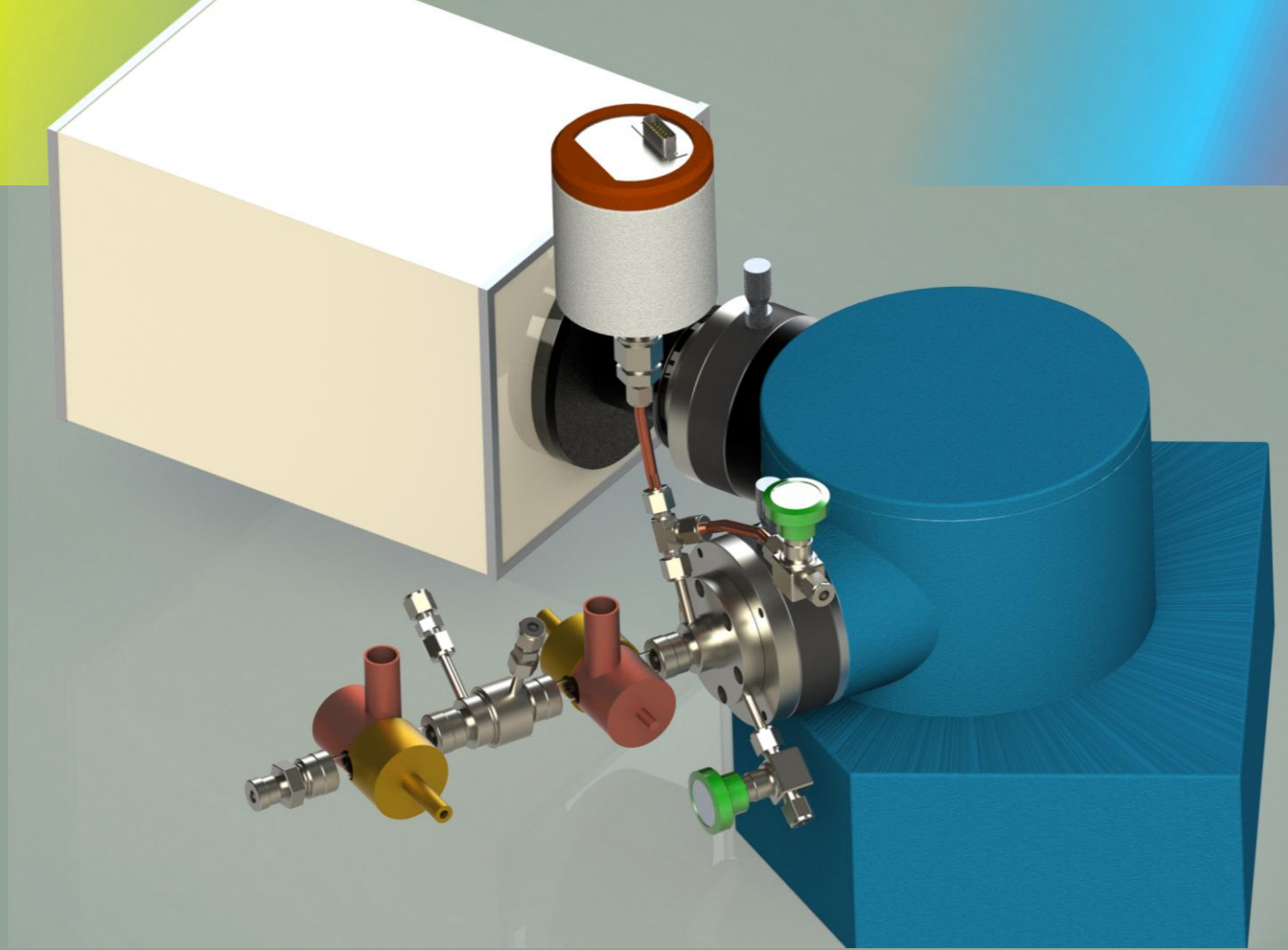
- Baratron pressure gauge
 - 1000 Torr
 - 10 Torr

VUV absorption system



➤ System Overview

- VM502
- Absorption cell
- PMT
- F2 lamp
- Pressure gauge



➤ Discharge Lamp

- Micro-wave
- F2 gas

➤ PMT

- TE cooled
- Visible range
- sodium salicylate

➤ VM502

- VUV grating 1200 l/mm
- 200mm focusing length
- Vacuum monochromator

Microwave discharge lamp and absorption cell

➤ Perspective view

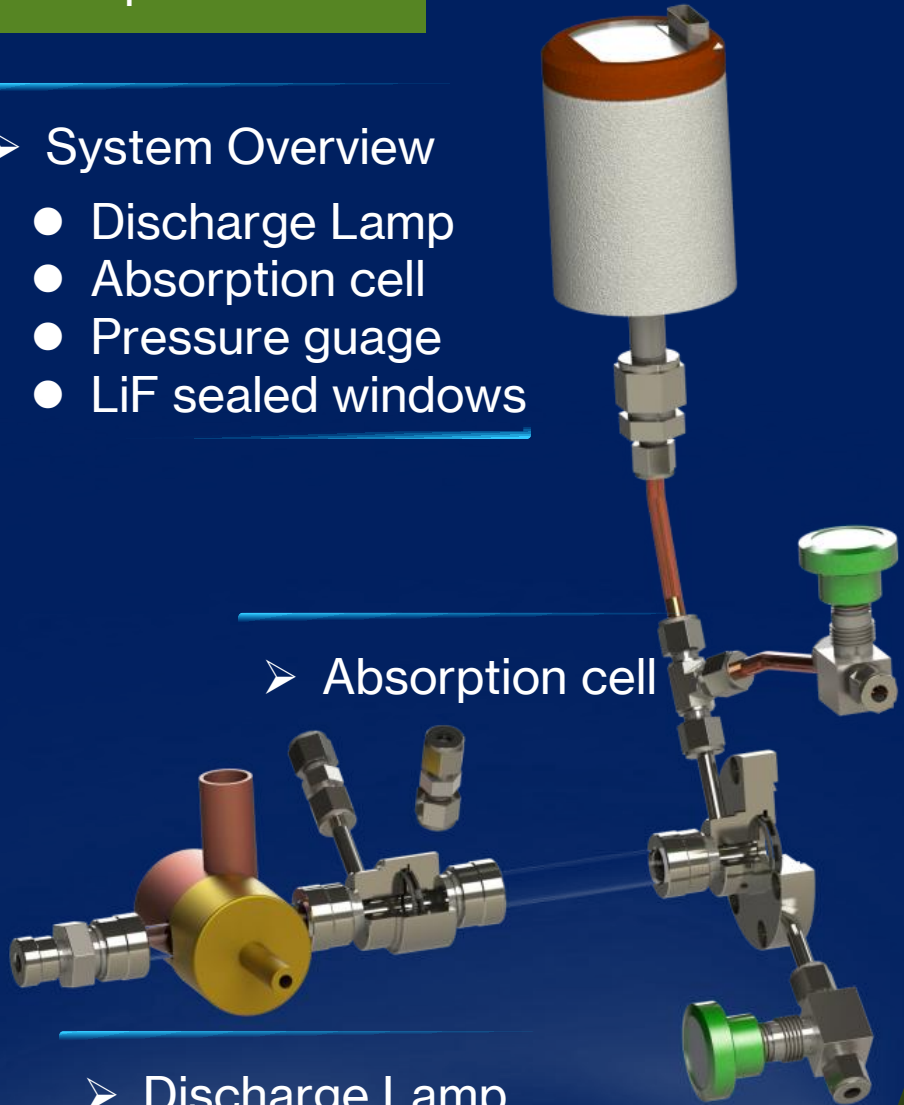
➤ System Overview

- Discharge Lamp
- Absorption cell
- Pressure guage
- LiF sealed windows

- Baratron pressure gauge

➤ Absorption cell

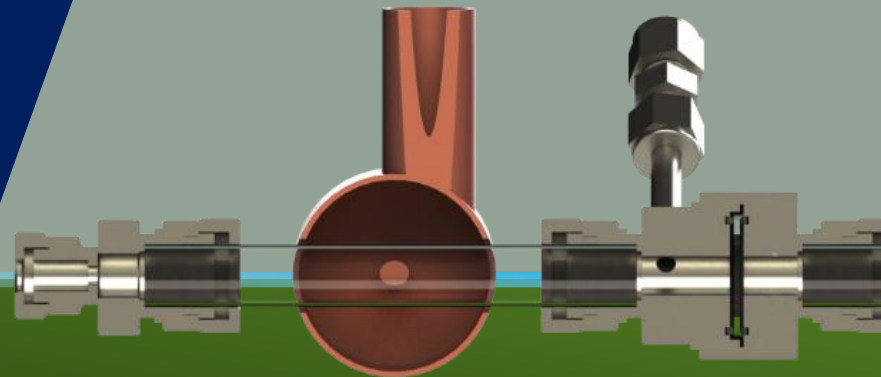
➤ Discharge Lamp



➤ Cross section view



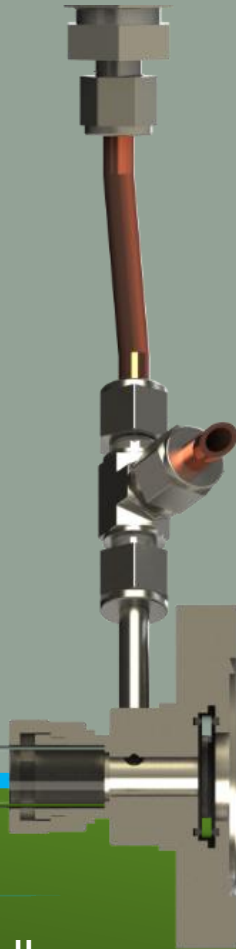
➤ Waveguide



➤ Discharge Lamp

- Micro-wave
- F2

➤ Absorption cell

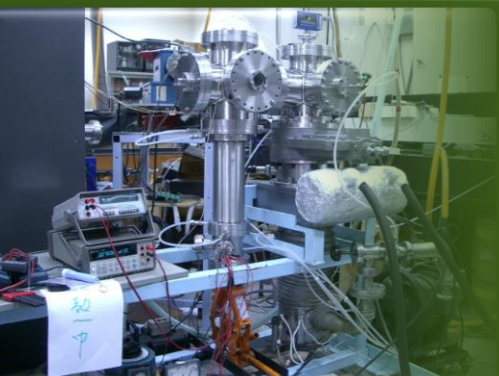
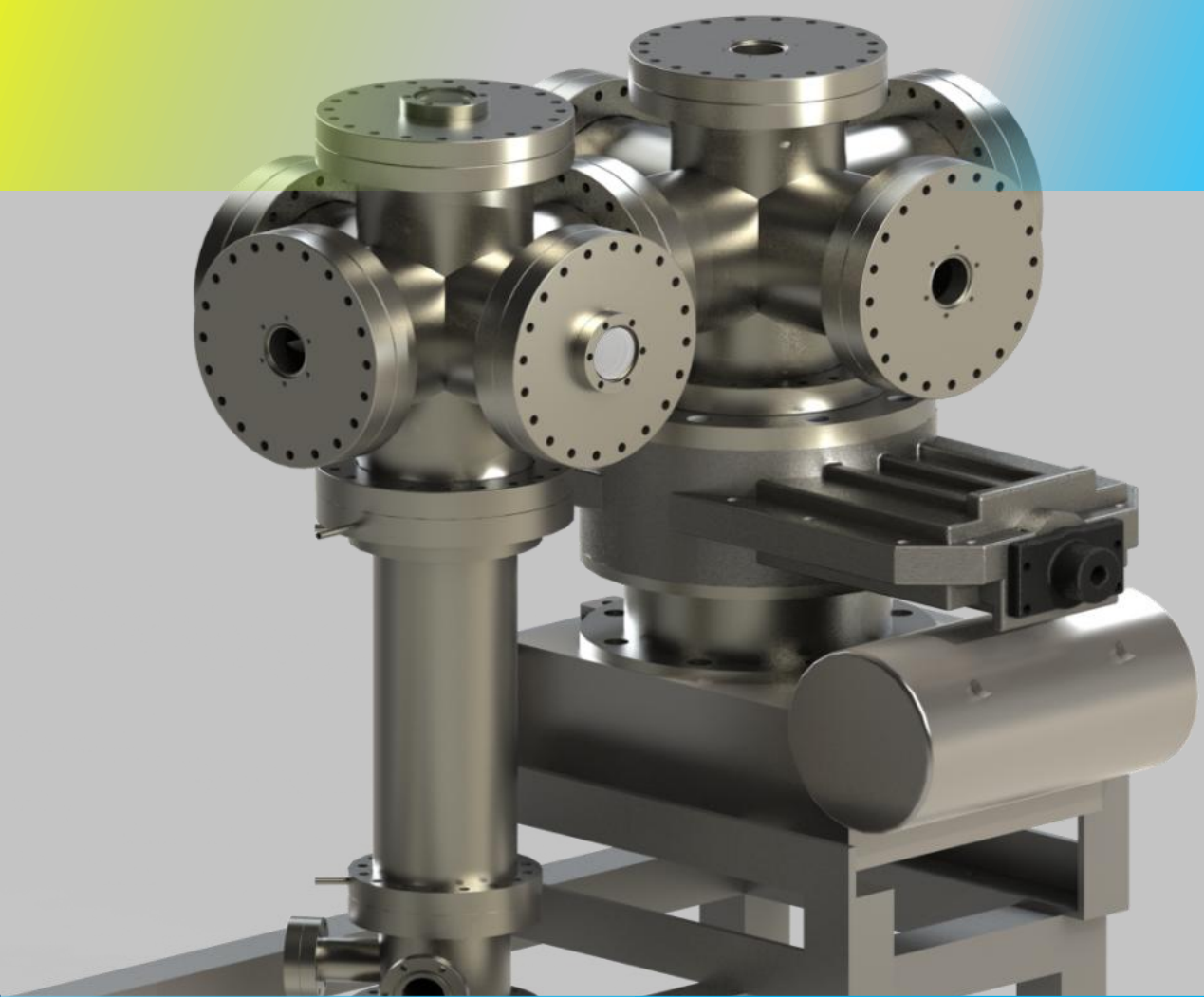


Stark effect system



➤ Experiences:

- BL04 Seya
- BL21B2
- Li, Na
- Mg, Ca, Sr, Ba



➤ Interaction zone

- 5 mm gap
- 25 KV
- 50 KV/cm

➤ Ion detector

- 2-stage MCP
- Amp up to 10^5

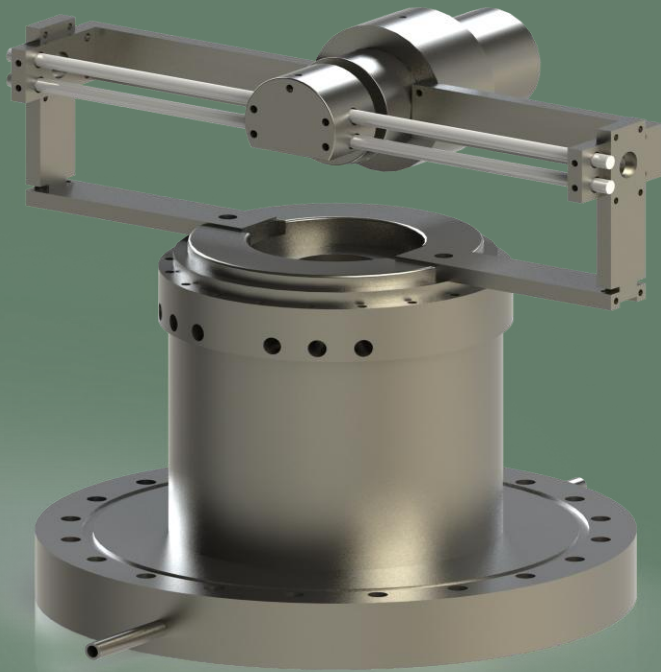
➤ Atom gun

- Temp 1200 C
- Vacuum oven
- Tungsten filament
- Water cooling

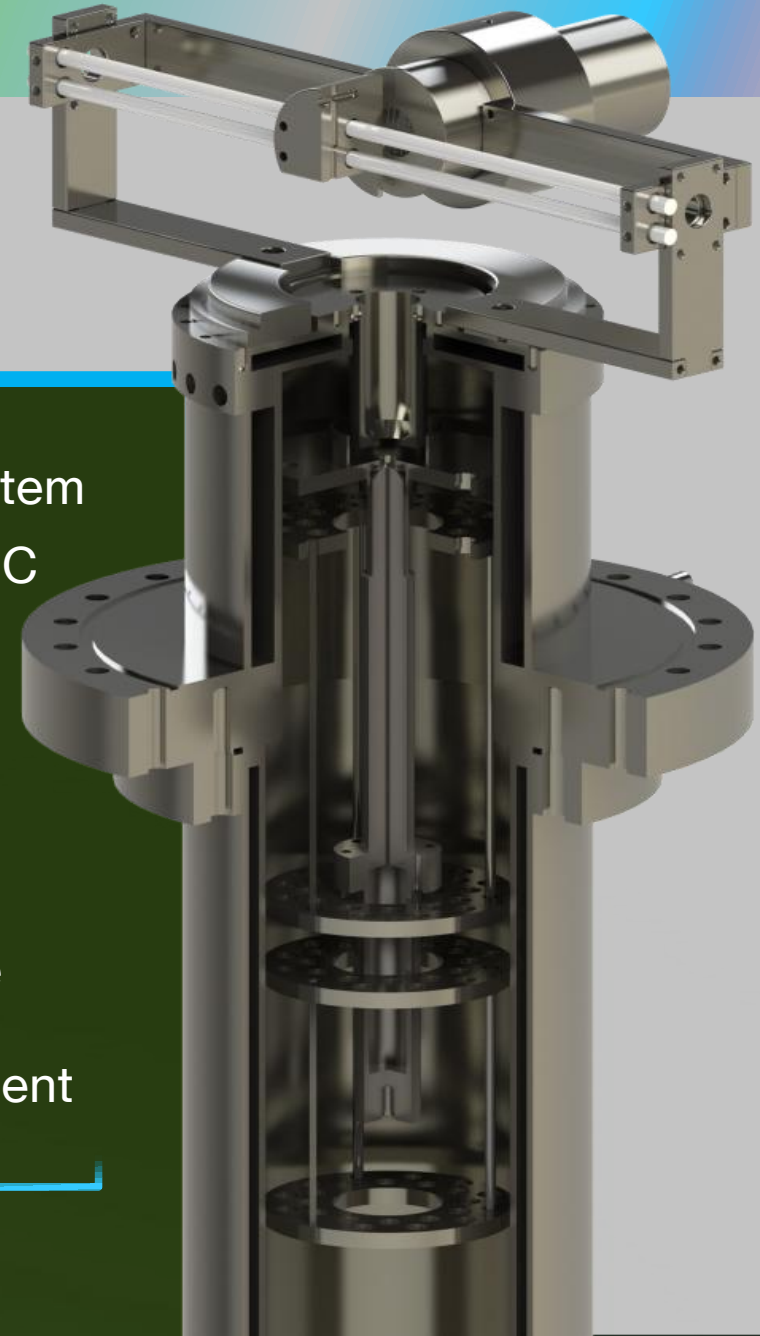
Atom gun system



➤ perspective view



➤ Cross section view



➤ sample preparation system

- Vacuum Oven 1200C
- Radiation shielding
- Cape Water cooling
- Main water cooling
- Atom beam shield
- Atom gun
- 200mm long nozzle
- 1x2 mm CS of nozzle
- Nozzle filament
- Atom container filament
- Interaction zone

➤ Interaction zone

- 5 mm gap
- 25 KV
- On the top of atom gun system

